

Transformation Code

```
WITH stats AS (  
  SELECT  
    'AGE' AS field_name,  
    AVG(AGE) AS mean,  
    STDDEV(AGE) AS stddev,  
    MIN(AGE) AS min,  
    MAX(AGE) AS max  
  FROM dataset  
  UNION ALL  
  SELECT  
    'SMOKING' AS field_name,  
    AVG(SMOKING) AS mean,  
    STDDEV(SMOKING) AS stddev,  
    MIN(SMOKING) AS min,  
    MAX(SMOKING) AS max  
  FROM dataset  
  UNION ALL  
  SELECT  
    'YELLOW_FINGERS' AS field_name,  
    AVG(YELLOW_FINGERS) AS mean,  
    STDDEV(YELLOW_FINGERS) AS stddev,  
    MIN(YELLOW_FINGERS) AS min,  
    MAX(YELLOW_FINGERS) AS max  
  FROM dataset  
  UNION ALL  
  SELECT  
    'ANXIETY' AS field_name,  
    AVG(ANXIETY) AS mean,  
    STDDEV(ANXIETY) AS stddev,  
    MIN(ANXIETY) AS min,  
    MAX(ANXIETY) AS max  
  FROM dataset  
  UNION ALL  
  SELECT  
    'PEER_PRESSURE' AS field_name,  
    AVG(PEER_PRESSURE) AS mean,  
    STDDEV(PEER_PRESSURE) AS stddev,  
    MIN(PEER_PRESSURE) AS min,  
    MAX(PEER_PRESSURE) AS max  
  FROM dataset
```



```
UNION ALL
SELECT
  'CHRONIC_DISEASE' AS field_name,
  AVG(CHRONIC_DISEASE) AS mean,
  STDDEV(CHRONIC_DISEASE) AS stddev,
  MIN(CHRONIC_DISEASE) AS min,
  MAX(CHRONIC_DISEASE) AS max
FROM dataset
UNION ALL
SELECT
  'FATIGUE' AS field_name,
  AVG(FATIGUE) AS mean,
  STDDEV(FATIGUE) AS stddev,
  MIN(FATIGUE) AS min,
  MAX(FATIGUE) AS max
FROM dataset
UNION ALL
SELECT
  'ALLERGY' AS field_name,
  AVG(ALLERGY) AS mean,
  STDDEV(ALLERGY) AS stddev,
  MIN(ALLERGY) AS min,
  MAX(ALLERGY) AS max
FROM dataset
UNION ALL
SELECT
  'WHEEZING' AS field_name,
  AVG(WHEEZING) AS mean,
  STDDEV(WHEEZING) AS stddev,
  MIN(WHEEZING) AS min,
  MAX(WHEEZING) AS max
FROM dataset
UNION ALL
SELECT
  'ALCOHOL_CONSUMING' AS field_name,
  AVG(ALCOHOL_CONSUMING) AS mean,
  STDDEV(ALCOHOL_CONSUMING) AS stddev,
  MIN(ALCOHOL_CONSUMING) AS min,
  MAX(ALCOHOL_CONSUMING) AS max
FROM dataset
UNION ALL
SELECT
  'COUGHING' AS field_name,
  AVG(COUGHING) AS mean,
  STDDEV(COUGHING) AS stddev,
  MIN(COUGHING) AS min,
  MAX(COUGHING) AS max
```



```
FROM dataset
UNION ALL
SELECT
  'SHORTNESS_OF_BREATH' AS field_name,
  AVG(SHORTNESS_OF_BREATH) AS mean,
  STDDEV(SHORTNESS_OF_BREATH) AS stddev,
  MIN(SHORTNESS_OF_BREATH) AS min,
  MAX(SHORTNESS_OF_BREATH) AS max
FROM dataset
UNION ALL
SELECT
  'SWALLOWING_DIFFICULTY' AS field_name,
  AVG(SWALLOWING_DIFFICULTY) AS mean,
  STDDEV(SWALLOWING_DIFFICULTY) AS stddev,
  MIN(SWALLOWING_DIFFICULTY) AS min,
  MAX(SWALLOWING_DIFFICULTY) AS max
FROM dataset
UNION ALL
SELECT
  'CHEST_PAIN' AS field_name,
  AVG(CHEST_PAIN) AS mean,
  STDDEV(CHEST_PAIN) AS stddev,
  MIN(CHEST_PAIN) AS min,
  MAX(CHEST_PAIN) AS max
FROM dataset
)
SELECT field_name, mean, stddev, min, max
FROM stats;
```

Derivation Dialog

user

[CONTEXT]

[TABLE dataset]

Column metadata:

GENDER (VARCHAR) ---- {'count': 3000, 'unique_count': 2, 'null_count': 0, 'sample_values': ['('M',)', '('F',)']}

AGE (BIGINT) ---- {'count': 3000, 'unique_count': 51, 'null_count': 0, 'sample_values': ['(41,)', '(46,)', '(56,)', '(30,)', '(65,)', '(67,)', '(72,)'']}

SMOKING (BIGINT) ---- {'count': 3000, 'unique_count': 2, 'null_count': 0, 'sample_values': ['(1,)', '(2,)'']}



YELLOW_FINGERS (BIGINT) ---- {'count': 3000, 'unique_count': 2, 'null_count': 0, 'sample_values': ['(1,)',
'(2,)']}

ANXIETY (BIGINT) ---- {'count': 3000, 'unique_count': 2, 'null_count': 0, 'sample_values': ['(1,)', '(2,)']}

PEER_PRESSURE (BIGINT) ---- {'count': 3000, 'unique_count': 2, 'null_count': 0, 'sample_values': ['(1,)',
'(2,)']}

CHRONIC_DISEASE (BIGINT) ---- {'count': 3000, 'unique_count': 2, 'null_count': 0, 'sample_values': ['(1,)',
'(2,)']}

FATIGUE (BIGINT) ---- {'count': 3000, 'unique_count': 2, 'null_count': 0, 'sample_values': ['(1,)', '(2,)']}

ALLERGY (BIGINT) ---- {'count': 3000, 'unique_count': 2, 'null_count': 0, 'sample_values': ['(1,)', '(2,)']}

WHEEZING (BIGINT) ---- {'count': 3000, 'unique_count': 2, 'null_count': 0, 'sample_values': ['(1,)', '(2,)']}

ALCOHOL_CONSUMING (BIGINT) ---- {'count': 3000, 'unique_count': 2, 'null_count': 0, 'sample_values':
['(1,)', '(2,)']}

COUGHING (BIGINT) ---- {'count': 3000, 'unique_count': 2, 'null_count': 0, 'sample_values': ['(1,)', '(2,)']}

SHORTNESS_OF_BREATH (BIGINT) ---- {'count': 3000, 'unique_count': 2, 'null_count': 0, 'sample_values':
['(1,)', '(2,)']}

SWALLOWING_DIFFICULTY (BIGINT) ---- {'count': 3000, 'unique_count': 2, 'null_count': 0, 'sample_values':
['(2,)', '(1,)']}

CHEST_PAIN (BIGINT) ---- {'count': 3000, 'unique_count': 2, 'null_count': 0, 'sample_values': ['(1,)', '(2,)']}

LUNG_CANCER (VARCHAR) ---- {'count': 3000, 'unique_count': 2, 'null_count': 0, 'sample_values': ["
('YES,)',", "('NO,)'"]}

Sample data:

| GENDER | AGE | SMOKING | YELLOW_FINGERS | ANXIETY | PEER_PRESSURE | CHRONIC_DISEASE |
FATIGUE | ALLERGY | WHEEZING | ALCOHOL_CONSUMING | COUGHING | SHORTNESS_OF_BREATH |
SWALLOWING_DIFFICULTY | CHEST_PAIN | LUNG_CANCER |
0| M | 65 | 1 | 1 | 1 | 2 | 2 | 1 | 2 | 2 | 2 | 2 | 2 | 1 | NO |
1| F | 55 | 1 | 2 | 2 | 1 | 1 | 2 | 2 | 2 | 1 | 1 | 1 | 2 | NO |
2| F | 78 | 2 | 2 | 1 | 1 | 1 | 2 | 1 | 2 | 1 | 1 | 2 | 1 | YES |
3| M | 60 | 2 | 1 | 1 | 1 | 2 | 1 | 2 | 1 | 1 | 2 | 1 | 2 | YES |
4| F | 80 | 1 | 1 | 2 | 1 | 1 | 2 | 1 | 2 | 1 | 1 | 1 | 1 | NO |

[GOAL]

Can you identify any outliers in the data?

[OUTPUT]

assistant

```
```json
{
 "mode": "summary",
 "recommendation": "To identify outliers, we can calculate summary statistics for numerical fields, such as mean, standard deviation, min, and max. This will help us understand the distribution of the data and potentially identify outliers.",
 "output_fields": ["field_name", "mean", "stddev", "min", "max"],
 "chart_type": "boxplot",
 "visualization_fields": ["field_name", "mean", "stddev"]
}
```
```

```
```sql
WITH stats AS (
 SELECT
 'AGE' AS field_name,
 AVG(AGE) AS mean,
 STDDEV(AGE) AS stddev,
 MIN(AGE) AS min,
 MAX(AGE) AS max
 FROM dataset
 UNION ALL
 SELECT
 'SMOKING' AS field_name,
 AVG(SMOKING) AS mean,
 STDDEV(SMOKING) AS stddev,
 MIN(SMOKING) AS min,
 MAX(SMOKING) AS max
 FROM dataset
 UNION ALL
 SELECT
 'YELLOW_FINGERS' AS field_name,
 AVG(YELLOW_FINGERS) AS mean,
```



```
STDDEV(YELLOW_FINGERS) AS stddev,
MIN(YELLOW_FINGERS) AS min,
MAX(YELLOW_FINGERS) AS max
FROM dataset
UNION ALL
SELECT
'ANXIETY' AS field_name,
AVG(ANXIETY) AS mean,
STDDEV(ANXIETY) AS stddev,
MIN(ANXIETY) AS min,
MAX(ANXIETY) AS max
FROM dataset
UNION ALL
SELECT
'PEER_PRESSURE' AS field_name,
AVG(PEER_PRESSURE) AS mean,
STDDEV(PEER_PRESSURE) AS stddev,
MIN(PEER_PRESSURE) AS min,
MAX(PEER_PRESSURE) AS max
FROM dataset
UNION ALL
SELECT
'CHRONIC_DISEASE' AS field_name,
AVG(CHRONIC_DISEASE) AS mean,
STDDEV(CHRONIC_DISEASE) AS stddev,
MIN(CHRONIC_DISEASE) AS min,
MAX(CHRONIC_DISEASE) AS max
FROM dataset
UNION ALL
SELECT
'FATIGUE' AS field_name,
AVG(FATIGUE) AS mean,
STDDEV(FATIGUE) AS stddev,
MIN(FATIGUE) AS min,
MAX(FATIGUE) AS max
FROM dataset
UNION ALL
SELECT
'ALLERGY' AS field_name,
```

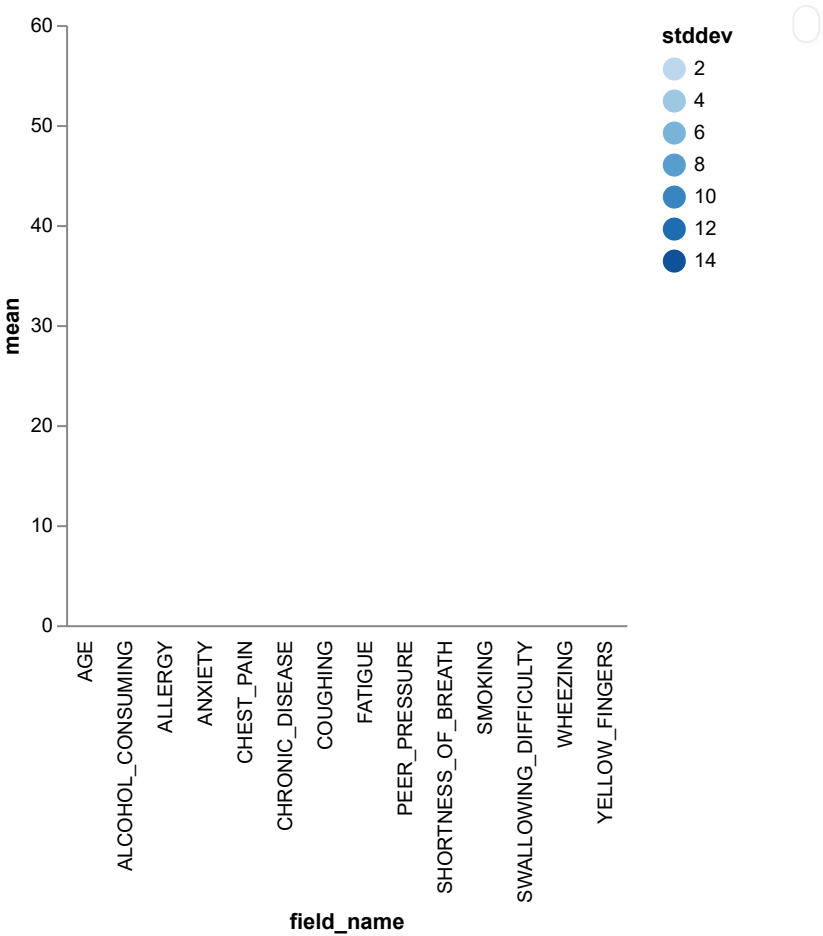


```
AVG(ALLERGY) AS mean,
STDDEV(ALLERGY) AS stddev,
MIN(ALLERGY) AS min,
MAX(ALLERGY) AS max
FROM dataset
UNION ALL
SELECT
'WHEEZING' AS field_name,
AVG(WHEEZING) AS mean,
STDDEV(WHEEZING) AS stddev,
MIN(WHEEZING) AS min,
MAX(WHEEZING) AS max
FROM dataset
UNION ALL
SELECT
'ALCOHOL_CONSUMING' AS field_name,
AVG(ALCOHOL_CONSUMING) AS mean,
STDDEV(ALCOHOL_CONSUMING) AS stddev,
MIN(ALCOHOL_CONSUMING) AS min,
MAX(ALCOHOL_CONSUMING) AS max
FROM dataset
UNION ALL
SELECT
'COUGHING' AS field_name,
AVG(COUGHING) AS mean,
STDDEV(COUGHING) AS stddev,
MIN(COUGHING) AS min,
MAX(COUGHING) AS max
FROM dataset
UNION ALL
SELECT
'SHORTNESS_OF_BREATH' AS field_name,
AVG(SHORTNESS_OF_BREATH) AS mean,
STDDEV(SHORTNESS_OF_BREATH) AS stddev,
MIN(SHORTNESS_OF_BREATH) AS min,
MAX(SHORTNESS_OF_BREATH) AS max
FROM dataset
UNION ALL
SELECT
```



```
'SWALLOWING_DIFFICULTY' AS field_name,
AVG(SWALLOWING_DIFFICULTY) AS mean,
STDDEV(SWALLOWING_DIFFICULTY) AS stddev,
MIN(SWALLOWING_DIFFICULTY) AS min,
MAX(SWALLOWING_DIFFICULTY) AS max
FROM dataset
UNION ALL
SELECT
 'CHEST_PAIN' AS field_name,
 AVG(CHEST_PAIN) AS mean,
 STDDEV(CHEST_PAIN) AS stddev,
 MIN(CHEST_PAIN) AS min,
 MAX(CHEST_PAIN) AS max
FROM dataset
)
SELECT field_name, mean, stddev, min, max
FROM stats;
...
```





AI generated results can be inaccurate, inspect it!



sample data from **view\_cjxn** (5 columns × 14 rows)

[show column stats](#)

field_name	mean	stddev	min	max
AGE	55.169	14.7237461911	30	80
SMOKING	1.491	0.5000023341	1	2
YELLOW_FINGERS	1.514	0.4998872831	1	2
ANXIETY	1.494	0.5000473469	1	2
PEER_PRESSURE	1.499	0.500082354	1	2
CHRONIC_DISEASE	1.5096666667	0.4999898854	1	2
FATIGUE	1.4896666667	0.4999765472	1	2
ALLERGY	1.5066666667	0.5000389003	1	2
WHEEZING	1.4973333333	0.5000762418	1	2
.....	.....	.....	.....	.....